The Web Application Hacker's Handbook Finding and Exploiting Security Flaws (2nd Edition)

by Dafydd Stuttard and Marcus Pinto (Wiley, 2011)

Comprehensive Book Summary

Chapter-by-chapter overview with tools, techniques, and key takeaways

Prepared as a reading companion — not a substitute for the full text.

About This Summary

This document summarizes all 21 chapters of *The Web Application Hacker's Handbook, 2nd Edition* by Dafydd Stuttard and Marcus Pinto — widely regarded as the definitive, encyclopedic reference on web application penetration testing. Unlike a lab-driven walkthrough, this book is organized as a deep, vulnerability-by-vulnerability catalogue: for each category of flaw it covers the underlying mechanism, how to find it, how to exploit it, and how to remediate it, closing with a master checklist methodology (Chapter 21).

Each chapter section below includes a synopsis, key techniques/concepts, and a box listing the **tools** the authors reference for that topic. Because this book emphasizes methodology and manual technique over automated tooling, far fewer named tools appear than in a pure lab manual — the authors explicitly focus on a small, proven toolkit (centered on their own tool, Burp Suite) rather than an exhaustive tool tour. Tools appearing for the **first time** are flagged in orange.

Book Structure (4 Logical Sections)

- **Context-setting** (Ch. 1–4): the state of web app security, core defense mechanisms, web technologies primer, and mapping an application's attack surface.
- **Attacking core defense mechanisms** (Ch. 5–8): client-side controls, authentication, session management, and access control.
- **Attacking specific vulnerability classes** (Ch. 9–18): data stores (SQL/NoSQL/XPath/LDAP injection), back-end components, application logic, XSS, other user-side attacks, automation, information disclosure, native code, application architecture, and the application server.
- **Pulling it together** (Ch. 19–21): source code review, the hacker's toolkit, and a comprehensive attack methodology/checklist.

Chapter 1: Web Application (In)security

Sets the stage: despite widespread "this site is secure" assurances, most web applications remain exploitable with modest skill. The chapter frames the single root cause behind nearly every web vulnerability: applications must accept arbitrary, untrusted input from users, and defects arise whenever that input is processed unsafely.

It also explains why web app flaws matter beyond the app itself — a compromised web app often gives an attacker a foothold into an organization's wider internal network, making the web application layer the modern "security perimeter."

Key techniques & concepts

- Core problem: "users can submit arbitrary input" — every other vulnerability class in the book is a variation on unsafe handling of that input.
- Web apps are now effectively the primary attack surface of most organizations, more so than traditional network perimeters.
- Introduces the book's dual purpose: attacking flaws AND understanding remediation advice.

Chapter 2: Core Defense Mechanisms

Describes the security mechanisms every application should implement to manage the fact that input is untrusted: handling user access (authentication, session management, access control), handling user input (boundary validation, multistep validation/canonicalization), and handling attackers (error handling, audit logging, alerting, reacting to attacks).

Also covers application management functions (admin interfaces) as part of the attack surface. Understanding these intended mechanisms is a prerequisite for the rest of the book, since every later chapter targets a specific defense mechanism failing.

Key techniques & concepts

- Three pillars of user-access control: authentication, session management, access control — each gets its own dedicated chapter later (6, 7, 8).
- Boundary validation (validating input exactly where it crosses a trust boundary) is presented as the most robust input-handling strategy versus blacklisting/sanitization alone.
- Canonicalization bugs arise when input is validated before being decoded/normalized, letting encoded attacks slip through.

Chapter 3: Web Application Technologies

A primer on the technologies needed to understand the rest of the book: the HTTP protocol (requests, responses, methods, headers, cookies, status codes), HTTPS, HTTP proxies and authentication, REST, server-side vs. client-side functionality (HTML, JavaScript, VBScript, and browser-extension technologies like Java applets, ActiveX, Silverlight, Flash).

Also covers state/session concepts and encoding schemes used throughout web apps and attacks: URL encoding, Unicode encoding, HTML encoding, Base64, and hex encoding — all of which reappear constantly as both attack and evasion techniques in later chapters.

Key techniques & concepts

- Encoding schemes (URL/Unicode/HTML/Base64/hex) are foundational to understanding filter bypass and payload delivery in nearly every later attack chapter.
- Distinguishes server-side functionality (processing) from client-side functionality (presentation), a distinction central to Chapter 5's client-side bypass attacks.
- HTTP proxies are introduced as the core technique for intercepting and manipulating traffic.

Tools referenced in this chapter

HTTP intercepting proxy (concept)			

Chapter 4: Mapping the Application

The first practical exercise against any target: enumerating content and functionality via automated web spidering, user-directed spidering, and discovering hidden content (backup files, directories, parameters) through techniques like brute-forcing and inference from naming conventions.

Covers analyzing the application to identify all entry points for user input, fingerprinting server-side technologies and functionality, and compiling everything into a map of the application's full attack surface before any actual attacking begins.

Key techniques & concepts

- Distinguishes "application pages" from "functional paths" — the same functionality can be reached via multiple URLs/parameters that must all be mapped.
- Hidden content discovery (predictable filenames, backup extensions, robots.txt, comments) often reveals forgotten, unauthenticated, or debug functionality.
- Server-side technology fingerprinting (headers, error pages, file extensions) informs which vulnerability classes are most likely relevant later.

Tools referenced in this chapter

Web spiders (automated crawling tools)	Intercepting proxy (manual/user-directed spidering)		

Chapter 5: Bypassing Client-Side Controls

Covers the first real vulnerability category: trusting the client. This includes data transmitted via the client that the server assumes is unmodified (hidden form fields, cookies, URL parameters, the Referer header, opaque/encoded data, and the ASP.NET ViewState), and client-side checks on user input (length limits, script-based validation, disabled form elements).

Also covers heavier client-side technologies: intercepting traffic from and decompiling browser extensions (Java applets, Flash, Silverlight, ActiveX), and attaching a debugger to analyze client-side logic directly.

Key techniques & concepts

- Golden rule: any control enforced purely on the client can always be bypassed, because the attacker fully controls their own browser/client.
- The ASP.NET ViewState is highlighted as a common opaque-but-decodable client-side data store.
- Decompiling Java/.NET browser extensions (e.g., with JAD or a .NET decompiler/Reflector) can reveal client-enforced business logic and hardcoded secrets.

Tools referenced in this chapter

Intercepting proxy	JAD (Java decompiler)	.NET decompiler / Reflector	Debugger (attaching to client components)

★ Newly introduced in this chapter: JAD decompiler, .NET Reflector

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Chapter 6: Attacking Authentication

A deep catalogue of authentication flaws: design flaws (bad passwords, brute-forcible login, verbose failure messages, vulnerable credential transmission, weak password-change/forgotten-password/"remember me"/impersonation functions, non-unique or predictable usernames, predictable initial passwords, insecure credential distribution) and implementation flaws (fail-open login logic, defects in multistage login mechanisms, insecure credential storage).

Closes with a thorough "Securing Authentication" section covering strong credentials, secretive credential handling, proper validation, brute-force prevention, and logging/alerting — serving as both an attacker's checklist and a defender's hardening guide.

Key techniques & concepts

- Multistage login mechanisms (common in banking apps) are singled out as introducing new logic flaws not present in simple single-step logins.
- Verbose failure/error messages (e.g., "invalid username" vs. "invalid password") enable username enumeration, a key precursor to brute-forcing.
- Account-recovery and "remember me" functions are frequently the weakest link, bypassing otherwise strong primary authentication.

Tools referenced in this chapter

Intercepting proxy	Burp Intruder (automated brute-forcing)	Hydra	

★ Newly introduced in this chapter: Hydra

Chapter 7: Attacking Session Management

Session tokens let a stateless HTTP protocol track a user across requests — and breaking this mechanism can let an attacker bypass login entirely. Covers weaknesses in token generation (meaningful/encoded tokens, predictable tokens, poorly encrypted tokens) and weaknesses in token handling (disclosure on the network or in logs, vulnerable token-to-session mapping, vulnerable session termination, liberal cookie scope enabling hijacking).

Concludes with securing session management: generating strong random tokens and protecting them throughout their lifecycle (secure cookie attributes, transmission, expiry).

Key techniques & concepts

- A broken session-management mechanism can be as serious as broken authentication, since it lets attackers masquerade as other users without credentials.
- Token predictability can be analyzed statistically — sampling many tokens to detect patterns (sequential counters, weak PRNGs, timestamp-based tokens).
- Session fixation and liberal cookie scope are highlighted as subtle but serious hijacking vectors.

Tools referenced in this chapter

Intercepting proxy	Burp Sequencer (token randomness analysis)		

★ Newly introduced in this chapter: Burp Sequencer

Chapter 8: Attacking Access Controls

Access control relies on authentication and session management working correctly, but adds a third, independently-failing layer: enforcing what an authenticated user is actually allowed to do. Covers common vulnerabilities: completely unprotected functionality, identifier-based functions (insecure direct object references), unprotected multistage functions, exposed static files, platform misconfiguration, and insecure access-control methods (e.g., trusting a hidden parameter for role).

Describes systematic testing: using different user accounts, testing multistage processes and limited-access scenarios, testing direct method access, and checking HTTP-method restrictions. Ends with a "multilayered privilege model" as the recommended defense.

Key techniques & concepts

- Insecure Direct Object Reference (IDOR) — manipulating an identifier (e.g., an account or order number) to access another user's data — is a central theme of this chapter.
- Multistage functions often enforce access control only at the first step, letting an attacker jump directly to a later step.
- A robust defense enforces access control centrally and consistently, not per-page/per-link.

Tools referenced in this chapter

Intercepting proxy (multi-account comparison testing)			

Chapter 9: Attacking Data Stores

A deep dive into SQL injection: bypassing logins via injection, exploiting basic vulnerabilities, injecting into different SQL statement types, finding injection bugs, fingerprinting the database, using the UNION operator, extracting data, bypassing filters, second-order SQL injection, and advanced exploitation (out-of-band channels, inference, time-delay/blind techniques) — all compared across MS-SQL, Oracle, and MySQL.

Also covers escalating a SQL injection into full database/OS compromise, using dedicated SQL exploitation tools (notably **sqlmap**), a SQL syntax/error quick-reference, and prevention (parameterized queries). Extends the same injection concepts to **NoSQL** (including MongoDB), **XPath injection**, and **LDAP injection**.

Key techniques & concepts

- Covers the full spectrum of SQL injection: in-band (UNION-based), error-based, blind (boolean/time-based), and out-of-band exfiltration.
- Database-specific differences (syntax, error messages, available functions/commands) are tabulated across MS-SQL, Oracle, and MySQL for practical fingerprinting.
- sqlmap is highlighted as the standard automated tool for detecting and exploiting SQL injection once a candidate parameter is identified.
- Parameterized queries (prepared statements) are presented as the definitive SQL injection fix.

Tools referenced in this chapter

Intercepting proxy	sqlmap	SQL exploitation tools (generic)	

★ Newly introduced in this chapter: sqlmap

Chapter 10: Attacking Back-End Components

Covers injection categories beyond SQL: OS command injection (via Perl and ASP examples), injection into dynamic code execution (eval-style functions), file path manipulation (path traversal and file inclusion vulnerabilities, local and remote), and injection into XML interpreters (including XML External Entity / XXE injection) and SOAP services.

Also covers injecting into back-end HTTP requests (server-side request forgery concepts, HTTP parameter injection/pollution) and injecting into mail services (e-mail header manipulation, SMTP command injection) — each with dedicated detection and prevention guidance.

Key techniques & concepts

- OS command injection often hides behind seemingly safe wrapper functions that shell out to the OS with unsanitized input.
- Path traversal ("../") and file inclusion (LFI/RFI) can expose arbitrary files or achieve remote code execution depending on the target language/configuration.
- XXE injection is flagged as a serious, often-overlooked vector for file disclosure and SSRF via XML parsers that resolve external entities.
- HTTP parameter pollution/injection can manipulate back-end requests the application itself makes to other services.

Tools referenced in this chapter

Intercepting proxy	Manual injection testing		

Chapter 11: Attacking Application Logic

Argues that logic flaws are among the hardest vulnerabilities to find automatically because they are unique to each application's business rules, not a generic signature like SQL injection. The chapter presents 12 real-world case studies of broken logic — from a predictable "security question" oracle, to a flawed password-change function, to manipulable checkout/insurance/banking workflows, bulk-discount abuse, and race conditions in

login logic.

From these examples, the authors distill a generalized set of tests for locating logic flaws in any application: look for faulty assumptions designers make about user behavior, input completeness, and timing.

Key techniques & concepts

- Logic flaws can be more damaging than classic injection bugs because they often require no special payload — just doing something in an unexpected order or with unexpected values.
- Race conditions (e.g., "racing against the login") exploit the gap between a check and its corresponding action.
- Testing for logic flaws requires understanding the business process, not just the code/syntax.

Chapter 12: Attacking Users: Cross-Site Scripting

A comprehensive treatment of XSS, described as the single most prevalent web vulnerability. Covers the three varieties — reflected, stored, and DOM-based XSS — with real-world examples and the range of payloads and delivery mechanisms attackers use.

Provides a detailed methodology for finding and exploiting each XSS variety, even in obscure or heavily filtered contexts, followed by prevention guidance (output encoding, context-aware escaping, DOM-based XSS defenses).

Key techniques & concepts

- Reflected XSS requires a crafted link/request per victim; stored XSS persists in the app and hits every subsequent visitor automatically — making stored XSS generally more severe.
- DOM-based XSS occurs purely in client-side JavaScript and may never touch the server, requiring different detection techniques than server-reflected payloads.
- The chapter emphasizes bypassing input/output filters and WAFs as a core skill for real-world XSS exploitation.

Tools referenced in this chapter

Intercepting proxy	Firebug (DOM/JS inspection)		

★ Newly introduced in this chapter: Firebug

Chapter 13: Attacking Users: Other Techniques

Covers attacks against users beyond classic XSS: inducing user actions via request forgery (CSRF) and UI redress (clickjacking), capturing data cross-domain via HTML/CSS injection and JavaScript hijacking, and attacks on the Same-Origin Policy (including with browser extensions and HTML5).

Also covers HTTP header injection, cookie injection/session fixation, open redirection, client-side SQL injection, local privacy attacks (persistent cookies, cached content, browsing history, HTML5 local storage), ActiveX control vulnerabilities, and purely browser-side attacks (keystroke logging, history/search theft, internal network port scanning, DNS rebinding, and browser exploitation frameworks).

Key techniques & concepts

- CSRF exploits the browser's automatic inclusion of credentials (cookies) in cross-site requests to perform unwanted actions as the victim.
- UI redress / clickjacking tricks users into clicking something different from what they perceive.
- Local privacy attacks target data a browser stores locally (cache, history, local storage) regardless of any single application's own vulnerabilities.
- Browser Exploitation Frameworks (e.g., BeEF) are referenced as a way to weaponize a hooked browser for man-in-the-middle-style attacks against the user.

Tools referenced in this chapter

Intercepting proxy	Browser exploitation framework (e.g., BeEF)		

Chapter 14: Automating Customized Attacks

Argues that because every application is different, effective web hacking requires custom automation rather than generic scanners. Covers enumerating valid identifiers, detecting "hits" in bulk responses, scripting attacks, and the authors' own example custom tool (JAttack) for harvesting useful data and fuzzing for common vulnerabilities at scale.

The centerpiece is a detailed walkthrough of using **Burp Intruder** to automate customized attacks (payload positions, payload types, attack types, response analysis), followed by overcoming common barriers to automation: defensive session-handling mechanisms and CAPTCHA controls.

Key techniques & concepts

- Manual testing of every parameter/request combination doesn't scale — automation tailored to the specific app is essential for thorough coverage.
- Burp Intruder is presented as the primary tool for building and running customized automated attacks (brute-force, fuzzing, enumeration) against any HTTP request.
- Session-handling rules and CAPTCHA-solving strategies are covered as practical barriers testers must engineer around during automation.

Tools referenced in this chapter

Burp Intruder	Custom scripts (JAttack example)		

★ Newly introduced in this chapter: Burp Intruder

Chapter 15: Exploiting Information Disclosure

Covers how applications leak information under active attack and how testers should continuously monitor for it: script error messages, stack traces, informative debug messages, and verbose server/database error messages that reveal internal workings.

Also covers using publicly available information, deliberately engineering informative error messages, gathering published information about the target, and using inference to deduce hidden details — closing with prevention:

generic error messages, protecting sensitive data, minimizing client-side leakage.

Key techniques & concepts

- Information disclosure should be actively monitored throughout every other attack phase, not treated as a separate, isolated test.
- Verbose stack traces and database error messages often reveal technology stack, file paths, and even query structure useful for further attacks (e.g., SQL injection).
- Deliberately triggering errors (malformed input, boundary values) is a technique for extracting more information than default behavior reveals.

Chapter 16: Attacking Native Compiled Applications

Covers memory-corruption vulnerability classes that can arise in native (C/C++) components behind a web application: stack overflows, heap overflows, and "off-by-one" vulnerabilities, plus techniques for detecting buffer overflow vulnerabilities from a web-facing perspective.

Also covers integer vulnerabilities (overflows and signedness errors) and format string vulnerabilities, with an emphasis on recognizing symptoms of these bugs through web application behavior rather than full binary exploit development (which the authors note is a topic deserving a dedicated book).

Key techniques & concepts

- Even though most of the book is about logic/HTTP-level flaws, native code components (e.g., custom ISAPI filters, image-processing libraries) can introduce classic memory-corruption bugs.
- Integer overflows and signedness errors can silently bypass length/bounds checks.
- Format string bugs arise when user input is passed directly as a format specifier to C-style formatting functions.

Chapter 17: Attacking Application Architecture

Examines architecture-level weaknesses: tiered architectures that fail to properly segregate components, letting a compromise in one tier cascade into full application compromise, and shared hosting / Application Service Provider environments (virtual hosting, shared application services).

Covers how to attack shared environments — where a flaw in one tenant application might compromise the underlying host or neighboring applications — and extends this to early "cloud computing" shared-hosting threat models, with corresponding securing guidance.

Key techniques & concepts

- Proper segregation between application tiers limits the blast radius of a single compromised component.
- Shared/cloud hosting introduces a new threat actor: other tenants on the same infrastructure, not just external attackers.
- Securing shared environments requires strict isolation controls at the platform level, not just the application level.

Chapter 18: Attacking the Application Server

Targets the web server/application server itself rather than the application logic: vulnerable server configuration (default credentials, default content, directory listings, dangerous WebDAV methods, the server being abused as an open proxy, misconfigured virtual hosting).

Also covers vulnerable server software (application framework flaws, memory management vulnerabilities, encoding/canonicalization bugs), techniques for finding web server flaws, securing server software, and the role and limits of Web Application Firewalls (WAFs).

Key techniques & concepts

- Default credentials and default/sample content are still among the most common real-world application-server compromises.
- WebDAV methods (PUT, MOVE, etc.) left enabled can allow direct file upload/modification on the server.
- Encoding/canonicalization bugs in the server platform (not just the application) can bypass access restrictions or filters.
- WAFs are presented as a useful mitigating layer but not a substitute for fixing underlying flaws.

Tools referenced in this chapter

Nikto / Wikto (server/CGI vulnerability scanning)			

★ Newly introduced in this chapter: Nikto/Wikto

Chapter 19: Finding Vulnerabilities in Source Code

A completely different approach from black-box probing: reviewing an application's actual source code. Compares black-box vs. white-box testing and lays out a code review methodology, then gives "signatures" to grep/search for across common vulnerability classes (XSS, SQL injection, path traversal, arbitrary redirection, OS command injection, backdoor passwords, native software bugs, and suspicious source code comments).

Provides language-by-language cheat sheets — Java, ASP.NET, PHP, Perl, and JavaScript — covering how to identify user-supplied data, session interaction, and potentially dangerous APIs in each, plus database code review (SQL injection patterns, dangerous function calls) and tools for code browsing.

Key techniques & concepts

- Source code review can find vulnerabilities difficult or slow to detect by probing a live app, and doesn't require the target's cooperation if source is otherwise obtainable.
- Each language has characteristic "dangerous APIs" (e.g., eval, exec, raw SQL concatenation) that serve as fast triage points during a review.
- A grep-based signature approach lets even testers with limited programming experience perform an effective, targeted code review.

Tools referenced in this chapter

Code browsing / IDE search tools	grep-style signature searching		

Chapter 20: A Web Application Hacker's Toolkit

Consolidates every tool the authors actually use in real engagements. Covers web browsers (Internet Explorer, Firefox, Chrome) and their role in testing, then **integrated testing suites** — intercepting-proxy-centered toolkits like **Burp Suite** — explaining how they work internally and a recommended testing workflow, plus alternatives to the intercepting proxy approach.

Covers standalone automated vulnerability scanners (what they detect, their inherent limitations, the technical challenges they face, and current products available at the time), then a section on "Other Tools": Wikto/Nikto, Firebug, Hydra, and writing custom scripts — closing the book's practical toolkit discussion.

Key techniques & concepts

- Burp Suite is the authors' central recommended toolkit (intercepting proxy + Repeater + Intruder + Sequencer + Scanner, among other components referenced throughout the book).
- Automated scanners are useful for breadth and speed but have well-documented limitations: they struggle with logic flaws, multistage processes, and context-dependent vulnerabilities — manual testing remains essential.
- The chapter is explicit that no fully automated tool can replace skilled manual testing for a thorough assessment.

Tools referenced in this chapter

Burp Suite (full suite: Proxy, Repeater, Intruder, Sequencer, Scanner)	Internet Explorer / Firefox / Chrome (with extensions)	Standalone vulnerability scanners (e.g., commercial DAST products)	Wikto / Nikto
Firebug	Hydra	Custom scripts	

★ Newly introduced in this chapter: Burp Suite (full toolkit)

Chapter 21: A Web Application Hacker's Methodology

The book's capstone: a comprehensive, ordered checklist that organizes every technique from Chapters 1–20 into a single, repeatable attack methodology. Structured around roughly a dozen major phases — mapping the application, testing client-side controls, testing authentication, testing session management, testing access controls, testing for input-based vulnerabilities, testing for function-specific input vulnerabilities, testing for logic flaws, testing for shared-hosting vulnerabilities, testing for application-server vulnerabilities, miscellaneous checks (DOM-based attacks, local privacy, SSL cipher strength, same-origin policy), and following up any information leakage.

Each phase is broken into granular numbered steps (e.g., "4.3 Test for Username Enumeration," "7.2 Test for SQL Injection") that function as a practical, print-and-use checklist for running a full, structured penetration test against any web application.

Key techniques & concepts

- Intended as a standalone, actionable checklist readers can use directly during a real engagement once they understand the underlying chapters.
- Ordered by logical task dependencies (e.g., understand the app before testing input; test authentication before access controls).

- Functions as the book's practical summary — effectively a condensed index of the entire text.

Master Tool Index (by first appearance)

Compared to lab-style books, *The Web Application Hacker's Handbook* deliberately keeps its named toolset small and deep rather than broad: the authors center almost everything around an intercepting proxy (Burp Suite, which Dafydd Stuttard also created) and manual technique, with a handful of specialist tools for specific tasks. The table below lists every named tool, the chapter(s) where it is covered, and its purpose.

Tool	Chapter(s)	Purpose
Intercepting proxy (general concept)	3, 4+	Core technique: intercept/modify HTTP(S) requests and responses
JAD (Java decompiler)	5	Decompiles Java applets to reveal client-side logic
.NET Reflector	5	Decompiles .NET browser-extension assemblies
Hydra	6, 20	Online brute-forcing of login forms/credentials
Burp Sequencer	7	Statistical analysis of session token randomness/predictability
sqlmap	9	Automated detection and exploitation of SQL injection
Firebug	12, 20	Browser dev-tools extension for inspecting DOM/JavaScript (XSS analysis)
Browser Exploitation Framework (e.g., BeEF)	13	"Hooks" a victim browser for ongoing MITM-style control
Burp Intruder	14, 20	Automates customized, high-volume fuzzing/brute-force/enumeration attacks
Nikto / Wikto	18, 20	Web/CGI server configuration and vulnerability scanning
Burp Suite (full toolkit: Proxy, Repeater, Intruder, Sequencer, Scanner)	20	The authors' primary integrated testing suite for manual + semi-automated testin
Standalone DAST vulnerability scanners (commercial)	20	Automated breadth scanning, with noted limitations
Custom scripts (e.g., JAttack)	14, 20	Bespoke automation tailored to a specific application's quirks

Final Notes

This handbook (2011, 2nd edition) is intentionally the opposite of a quick-start lab manual: it is an encyclopedic, vulnerability-by-vulnerability reference meant to be read deeply and then used as an ongoing desk reference, with Chapter 21's methodology serving as the practical, reusable checklist. The authors explicitly scope the book to *web application* attacks only — network penetration testing, firewall attacks, and general server compromise are deliberately out of scope.

Because the book predates widespread HTML5/SPA/API-first architectures and modern frameworks, some specific technology references (ActiveX, Silverlight, classic ASP) are dated. However, the core vulnerability classes it catalogues — SQL/NoSQL/XPath/LDAP injection, XSS, CSRF, authentication and session-management flaws, access control failures, logic flaws, and server misconfiguration — remain the backbone of web application security testing today, which is why this book is still widely considered the field's standard reference.